Tuple Calculus

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It is a non-procedural query language. It specifies what you want, but not how to get it.
It uses first-order logic and predicate calculus.

Basic formulae

P, Q are predicates.

↳ A statement that can be true/false.
(expression of 1/more variables)

P∨Q, P∧Q, ¬P, P→Q, ∀P, ∃P

Format of TRC queries:

$\{T \mid P(T)\}$

T: tuple variable
P: formula over tuple variable T

↳ Retrieves the set of tuples T that satisfy P(T)

E.g. $\{T \mid T \in \text{Student} \wedge T.\text{age} > 20\}$

Tuple variables are of 2 types:
- bounded: these are variables bounded by quantifiers (∃, ∀);
- free: these are not bounded by quantifiers; universal variables;

free → E.g. $\exists T_1 \in \text{Student}$ $\{T \mid \dots\}$

Q. Suppliers(sid, rating), Parts(pid, pname, color), Catalog(sid, pid, cost).
Retrieve sids of suppliers who supply some red parts.

→ $\{T \mid \exists T_1 \in \text{Catalogue}\ \exists T_2 \in \text{Parts}\ (T_1.\text{pid} = T_2.\text{pid} \wedge T_2.\text{color} = \text{red} \wedge T = T_1.\text{sid})\}$

$\pi_{\text{sid}}\left(\sigma_{\substack{\text{catalogue.pid} \\ = \text{parts.pid}}}\left(\text{Catalogue} \times \sigma_{\substack{\text{color} \\ = \text{red}}}(\text{Parts})\right)\right) \equiv \pi_{\text{sid}}\left(\text{Catalogue} \bowtie_{\substack{\text{catalogue.pid} \\ = \text{parts.pid}}} \sigma_{\substack{\text{color} \\ = \text{red}}}(\text{Parts})\right)$

Tuple relational calculus uses existential quantifiers the way we use joins and conditional joins in relational algebra.

Q. Retrieve sids of suppliers who supplied at least 2 parts.

→ There are atleast 2 products with the same sids but diff pids.

$$\Pi_{sid}\left(catalogue \underset{\substack{catalogue.sid = C.sid \\ catalogue.pid \neq C.pid}}{\bowtie} \rho(C, catalogue)\right)$$

$$\{T \mid \exists T_1 \in catalogue, \exists T_2 \in catalogue \left(T_1.sid = T_2.sid \wedge T_1.pid \neq T_2.pid \wedge T = T_1.sid\right)\}$$

$$R \cup S \Leftrightarrow \{T \mid T \in R \vee T \in S\}$$

$$R \cap S \Leftrightarrow \{T \mid T \in R \wedge T \in S\}$$

$$R - S \Leftrightarrow \{T \mid T \in R \wedge \underset{\neg(T \in S)}{T \notin S}\}$$

Q. Suppliers(sid, rating), Parts(pid, pname, color), Catalog(sid, pid, cost).
Retrieve supplier ids with maximum rating.

→ max rating ≡ ≥ every
≡ ∪ − < some

$$\Pi_{sid}(supplier) - \Pi_{sid}\left(Supplier \underset{\substack{supplier.rating \\ < S.rating}}{\bowtie} \rho(S, supplier)\right)$$

$$\{T \mid \exists T_1 \in supplier (T = T_1.sid) \wedge \neg \exists T_1 \in supplier \exists T_2 \in Supplier (T_1.rating < T_2.rating)\}$$

Unsafe:
TRC queries with infinite records in the result are unsafe.

$$\{T \mid \neg T \in Supplier\}$$

Expressive power:

$$\left\{\begin{matrix}\text{Basic RA} \\ \text{queries'} \\ \text{expressive power}\end{matrix}\right\} \equiv \left\{\begin{matrix}\text{Safe TRC queries'} \\ \text{expressive power}\end{matrix}\right\}$$

Safe TRC or relational algebra cannot do the following:
- count of records or attribute values
- sum of attribute values
- average of attribute values